



Are You In Suspense?

Are you wondering what happened to your order or payment, even after a fortnight? But did you mention your name and address clearly while sending payment, and what the payment was for?

We get several payments every month without adequate instructions or even identity of the person sending the payment. As a result we are unable to take appropriate action on these, till we get a complaint - if at all.

At EFY, we want you to be our satisfied customer. So please do let us know in case your payment or instructions are not even acknowledged within a fortnight. We are not that lazy!

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Technology Drives Us

so that if one service is denied for any reason, they can fall back on other systems. IRNSS is almost available, too. Reliance on IRNSS is going to be much higher in India."

Design challenges

Once we start reducing the size of the GNSS receiver, we have to keep in mind several factors. For example, when we have several transistors within a space of less than a centimetre, interference from adjoining transmitters becomes an issue at some point. Magnetisation, current induced due to magnetisation, eddy current and hysteresis are some other factors that arise from the several tracks running around. Requirement of a proper heat-sink is also a major issue inside the chip.

Now, these devices have to be economically viable to make but also be affordable. So cost-reduction measures include low-cost oscillators and low-cost antennae for receivers.

For low-cost antennae, Planar Inverted F Antenna (PIFA) has been a very suitable substitute with very few drawbacks. To further lower cost, chips are being implemented on single-wafer silicon.

Market and manufacturers

"As per the GNSS market report 2014 of European GNSS agency, around the globe 3.6 billion GNSS devices are in use. By 2019, this is forecast to increase to over seven billion," says M.N. Vidyashankar, president, IESA.

That seems to be a very satisfying number for manufacturing companies, since GNSS is being used in different environments. We have GNSS embedded in smart devices, digital cameras, portable equipment, fitness gear and many more.

Vidyashankar adds, "GNSS is also used in various platforms that are used in automobiles and air navigation systems to display moving maps, information about location, speed, direction and nearby streets. Similarly, heavy equipment employ

GNSS in construction, mining and precision agriculture. Cyclists often use GNSS in racing and touring. Spacecraft also use GNSS as a navigational tool." A very wide market indeed!

Asian companies holding majority of handset revenues dominate the location based services segment. Among others, Furuno, Broadcom, Mediatek, ST Microelectronics, Qualcomm, u-Blox and VectorNav are some giants producing GNSS chips. Navika Electronics is an Indian fabless semiconductor company designing and manufacturing GNSS chipsets.

In our own backyard

GPS-GLONASS-GAGAN-IRNSS (G3I) is one of the latest products from Accord Software. This credit-card-size GNSS receiver can be used as a standalone device as well as in reliable timing solutions. The level of accuracy provided by G3I finds uses in domains of telecom, power distribution and industrial segments.

Developments in technology

Due to a large market and multiple GNSS chip providers, we have newer and better products coming out on a regular basis. These newer products, either in the form of firmware changes or hardware improvements, help in improving the current guidance systems by achieving certain changes.

Multiple receivers. In a recent release, VN-360 GPS-compass heading-and-position sensor was unveiled by VectorNav Technologies. It is an OEM GPS-compass module that provides an accurate, true North-heading solution for system integrators seeking a reliable alternative to magnetic based sensors. It is a miniature, cost-effective GPS based alternative that is unaffected by magnetic disturbances and changes to the magnetic environment.

Incorporating two onboard GNSS receivers, VN-360 calculates the